

Beyond GDP: what really drives national happiness?

BEE2041 Data Science in Economics, empirical project, April 2026

When the World Happiness Report (WHR) crowns Finland the happiest country on earth for the seventh year running, two questions follow naturally. First, *how* do we know? Second, *why* this country and not another? National wealth is the obvious suspect, but Afghanistan sits at the bottom of the ranking below countries with a twentieth of its income, so money cannot be the whole story.

The puzzle is older than the data. Richard Easterlin (1974) noticed that as post-war America roughly tripled its GDP per capita, the share of Americans calling themselves "very happy" barely moved. Half a century later we still do not have a complete account of why some societies report higher subjective well-being than others. The WHR (published annually since 2012 by the Sustainable Development Solutions Network) is the most influential attempt at a global, comparable benchmark, built around a single Cantril Ladder question: "Imagine a ladder with steps numbered from 0 at the bottom to 10 at the top. Suppose we say that the top of the ladder represents the best possible life for you and the bottom of the ladder represents the worst possible life for you. On which step would you say you personally feel you stand at this time?"

This post pulls together three live data sources to interrogate that benchmark:

1. Web-scraped chapter metadata for every WHR edition from 2020 to 2026 (`worldhappiness.report/ed/{year}/`).
2. The official WHR ranking spreadsheets for the same period (`files.worldhappiness.report/`).
3. World-Bank cross-country indicators retrieved through the `wbgapi` Python client.

I integrate the three in a small SQLite database, fit a sequence of OLS regressions, and finally use a causal forest to ask which countries see the biggest happiness premium from being above the income median. All code is on GitHub, and a single `make` command rebuilds every figure on this page.

How the data were assembled

The three sources are different in kind (a scrape, a bulk download, a REST API), and each one borrows directly from a piece of the BEE2041 toolkit:

Step	Tool	Where I got it
Scraping HTML	<code>requests</code> + <code>BeautifulSoup</code>	<i>Workflow, Modelling & Webscraping</i> lecture
Downloading the WHR <code>.xls</code> files	<code>requests</code> + <code>pandas.read_excel</code>	<i>Python for Data Management</i>
World-Bank pull	<code>wbgapi</code>	the data-wrangling problem set
Integration	<code>sqlite3</code> + a SQL JOIN	<i>Relational Database Management Systems</i>
Causal forest	<code>econml.dml.CausalForestDML</code>	Clarke's <code>immigrantEffects.py</code>

The scraper caches every page to `data/raw/html_cache/`, so once it has run once the rest of the build works offline.

```
In [1]: import pandas as pd, numpy as np, sqlite3
from IPython.display import Image, display

CLN = "data/clean/"
TAB = "output/tables/"
FIG = "output/figures/"

df = pd.read_csv(CLN + "analysis.csv")
print(f"Cross-section: {df.shape[0]} countries, {df.shape[1]} variables")
df.head(5)
```

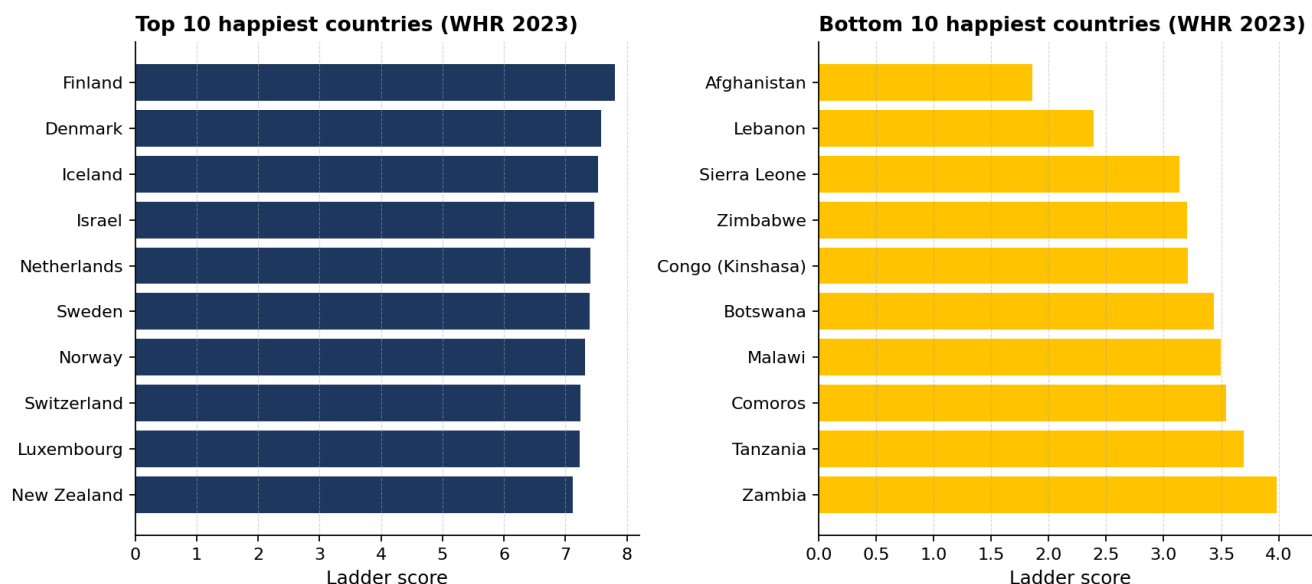
Cross-section: 136 countries, 16 variables

	country_code	country_name	year	ladder	log_gdp	social_support	life_exp_healthy	free
0	FIN	Finland	2023	7.8042	10.792010	0.968770	71.149994	0.96
1	DNK	Denmark	2023	7.5864	10.962164	0.954112	71.250145	0.93
2	ISL	Iceland	2023	7.5296	10.895531	0.982533	72.050018	0.93
3	ISR	Israel	2023	7.4729	10.638705	0.943344	72.697205	0.80
4	NLD	Netherlands	2023	7.4030	10.942279	0.930499	71.550018	0.88

The headlines, scraped and ranked

Before doing any modelling, it pays to look at the raw numbers. Finland tops the table with a Ladder score of 7.8 against Afghanistan's 1.9. That is a six-point gap on a ten-point scale, which is a *huge* gap for any explanation to close.

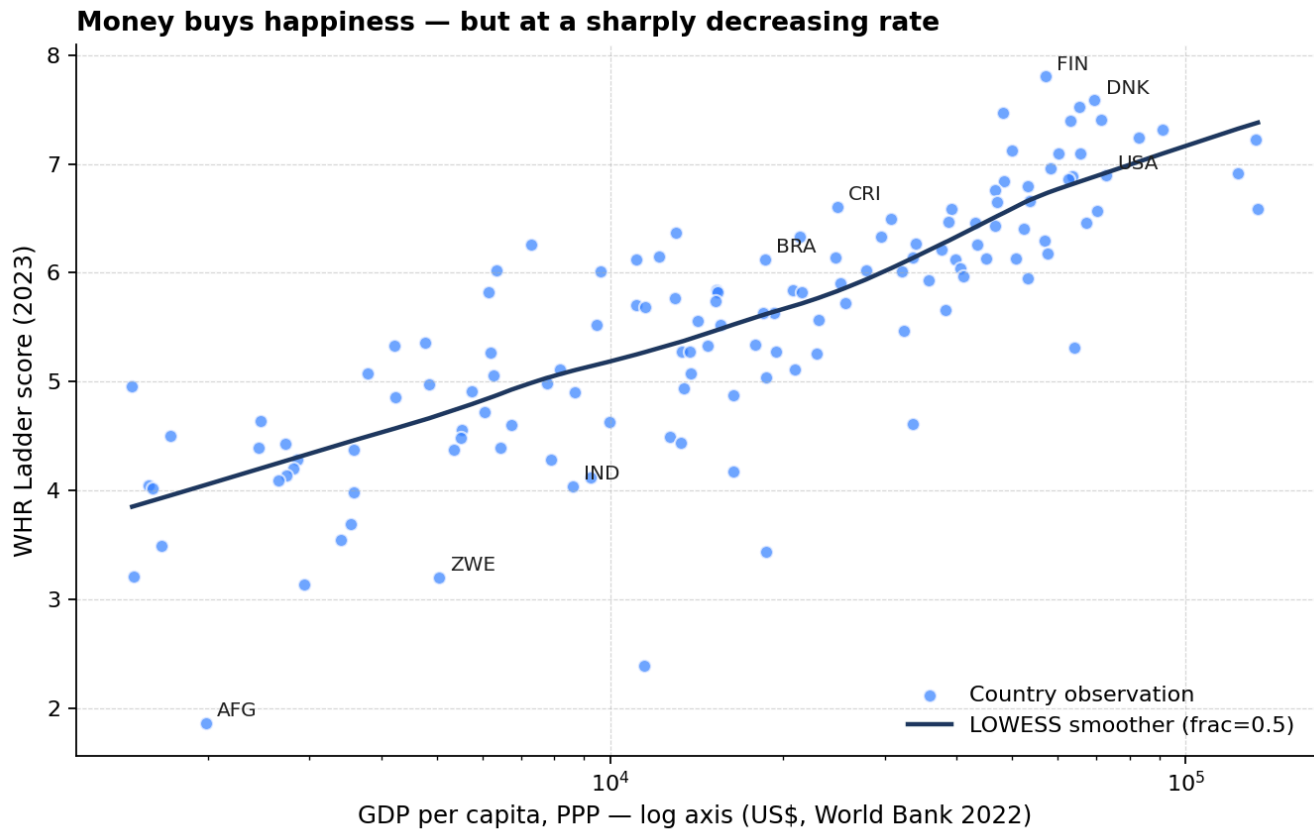
```
In [2]: display(Image(filename=FIG + "top_bottom_10.png"))
```



Money matters, but with diminishing returns

Plotting the Ladder against GDP per capita on a log axis (the usual choice, since income enters utility logarithmically) gives a clear upward relationship, but the LOWESS smoother flattens noticeably above roughly \$30,000 PPP. Costa Rica out-ranks the United States despite earning less than half of US income, a small-scale version of the Easterlin paradox.

```
In [3]: display(Image(filename=FIG + "gdp_vs_ladder.png"))
```

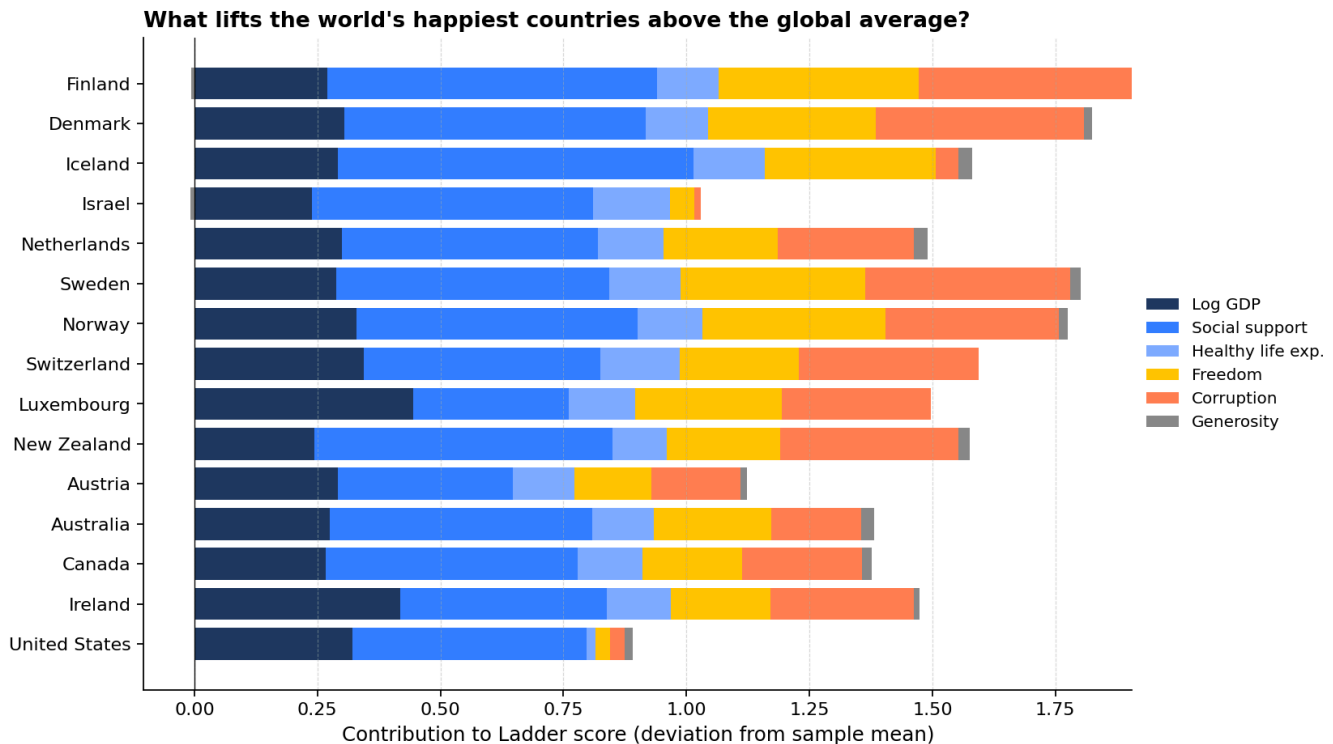


Two features complicate the simple "money buys happiness" reading. The curve clearly bends: doubling income from 2,000 to 4,000 PPP buys roughly the same Ladder-point increase as doubling from 32,000 to 64,000, which is what we would expect from logarithmic utility. And the vertical scatter is wide; countries with similar income levels routinely differ by close to two Ladder points. That residual variation is where the rest of this post lives.

Decomposing the Ladder: what carries the rich-country premium?

Alongside the Ladder score the WHR ships six explanatory variables: log GDP per capita, social support, healthy life expectancy, freedom to make life choices, perceptions of corruption, and generosity. To see how each one contributes I fit a plain OLS regression of the Ladder on these six, multiply each country's covariate (centred on the sample mean) by the fitted coefficient, and stack the contributions. The chart below shows the breakdown for the top-15 happiest countries.

```
In [4]: display(Image(filename=FIG + "decomposition.png"))
```



Two things stand out. First, *social support* (the share of respondents who say they have someone to count on in times of trouble) is the single biggest contributor in almost every Nordic country, with income only third. Second, *freedom* is large and stable across the whole top-15; the highest-ranked country with low freedom is Israel (4th), and it makes up the difference through unusually high social support and life expectancy.

The wider implication is that the gap between the world average and Finland's score is mostly *not* bought with money. Trust in institutions and in other people does most of the work, which lines up with Robert Putnam's social-capital research and Bo Rothstein's account of the Nordic "high-trust equilibrium".

OLS, six ways

Working in the same style as the BEE2041 wrangling problem set, I fit six specifications, adding one block of regressors at a time. Robust (HC3) standard errors throughout; the full LaTeX-formatted table is in `output/tables/regression_table.tex`.

```
In [5]: reg = pd.read_csv(TAB + "regression_summary.csv")
piv = (reg[reg.term != "const"]
       .pivot(index="term", columns="model", values="coef")
       .round(3)
       .fillna("-"))
fit = reg[["model", "r2", "n"]].drop_duplicates().set_index("model")
fit_summary = pd.DataFrame({
    "R²": fit["r2"].round(3).astype(str),
    "N": fit["n"].astype(int).astype(str),
}).T
```

```
display(piv)
display(fit_summary)
```

model	m1	m2	m3	m4	m5	m6
term						
corruption	—	—	—	-0.762	-0.764	-0.118
freedom	—	—	—	2.487	2.486	0.244
life_exp_healthy	—	0.035	0.015	0.011	0.013	0.056
log_gdp	0.797	0.644	0.329	0.27	—	0.274
log_gdp_wb	—	—	—	—	0.272	—
social_support	—	—	4.606	3.671	3.612	0.417

model	m1	m2	m3	m4	m5	m6
R ²	0.653	0.661	0.766	0.831	0.831	0.831
N	134	134	134	134	134	134

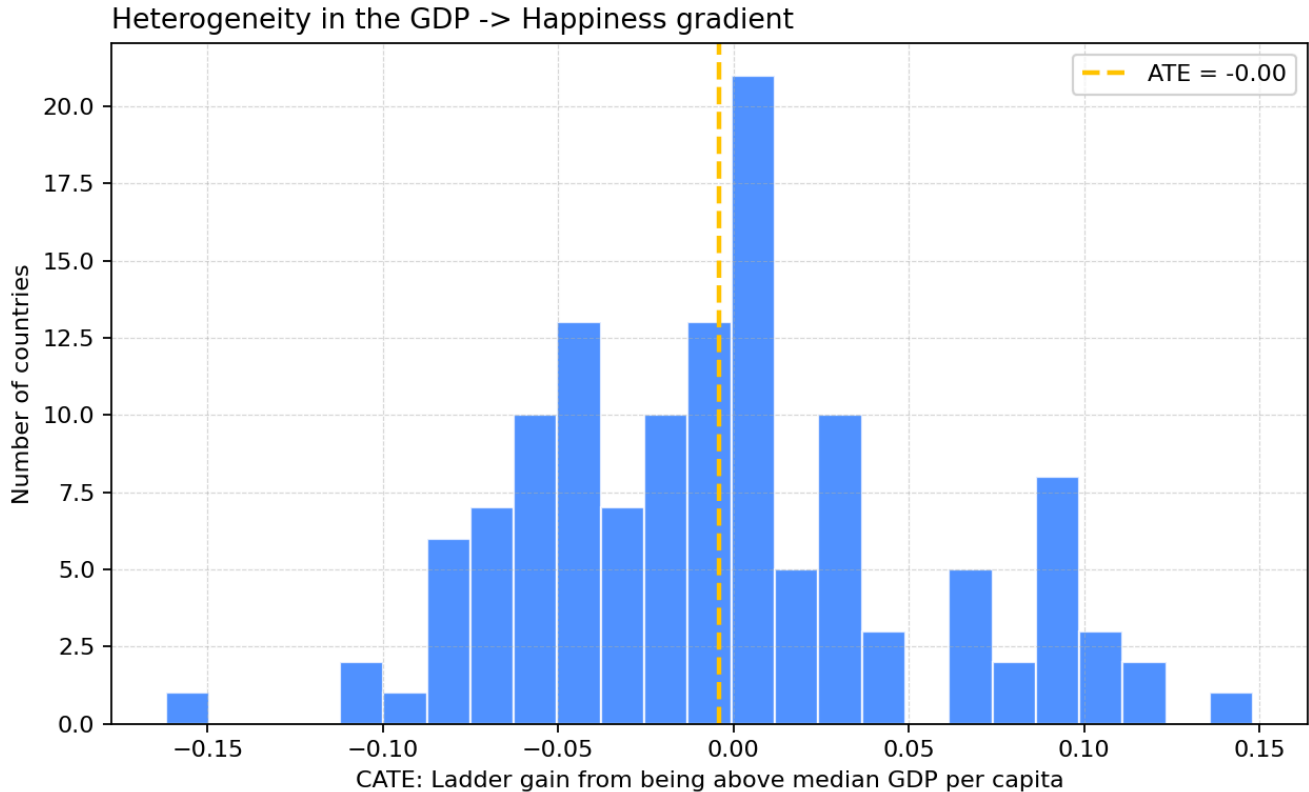
A few things to notice. The log-GDP coefficient falls from about 0.80 on its own to 0.27 once social support, freedom and corruption are controlled for: most of the apparent income premium is mediated through those other channels. Social support, freedom and corruption stay significant in every column where they appear, and in the standardised specification (m6) social support has the largest absolute coefficient (0.42). It is also reassuring that the two independent log-GDP measures (the WHR's bundled value and the World Bank's PPP series) give nearly identical estimates: a sanity check that the data are not introducing strange artefacts.

A causal forest: when does income still buy happiness?

Cross-country comparisons are obviously not a randomised trial; rich and poor countries differ on dozens of margins at once. To probe heterogeneity in a flexible, non-parametric way I fit a causal forest (Wager & Athey, 2018; Athey, Tibshirani & Wager, 2019) with `econml.dml.CausalForestDML`, following the same recipe Damian uses in `immigrantEffects.py`.

I encode the "treatment" as a binary indicator for being above the sample-median GDP per capita PPP. The conditioning set X has six potential moderators: healthy life expectancy, social support, freedom, corruption, internet penetration and urban share. The CATE we recover should be read as *descriptive*: the conditional difference in Ladder score between high- and low-income countries with the same values of the six moderators.

```
In [6]: display(Image(filename=FIG + "cate_hist.png"))
```



```
In [7]: import json
with open(TAB + "cate_headline.json") as f: hdl = json.load(f)
print(f"ATE = {hdl['ate']:+.3f} (95% CI: [{hdl['ate_ci'][0]:+.3f}, {hdl['ate_ci'][1]:+.3f}])")
print(f"CATEs: min = {hdl['cate_min']:+.2f}, median = {hdl['cate_med']:+.2f}, max = {hdl['cate_max']:+.2f}")
```

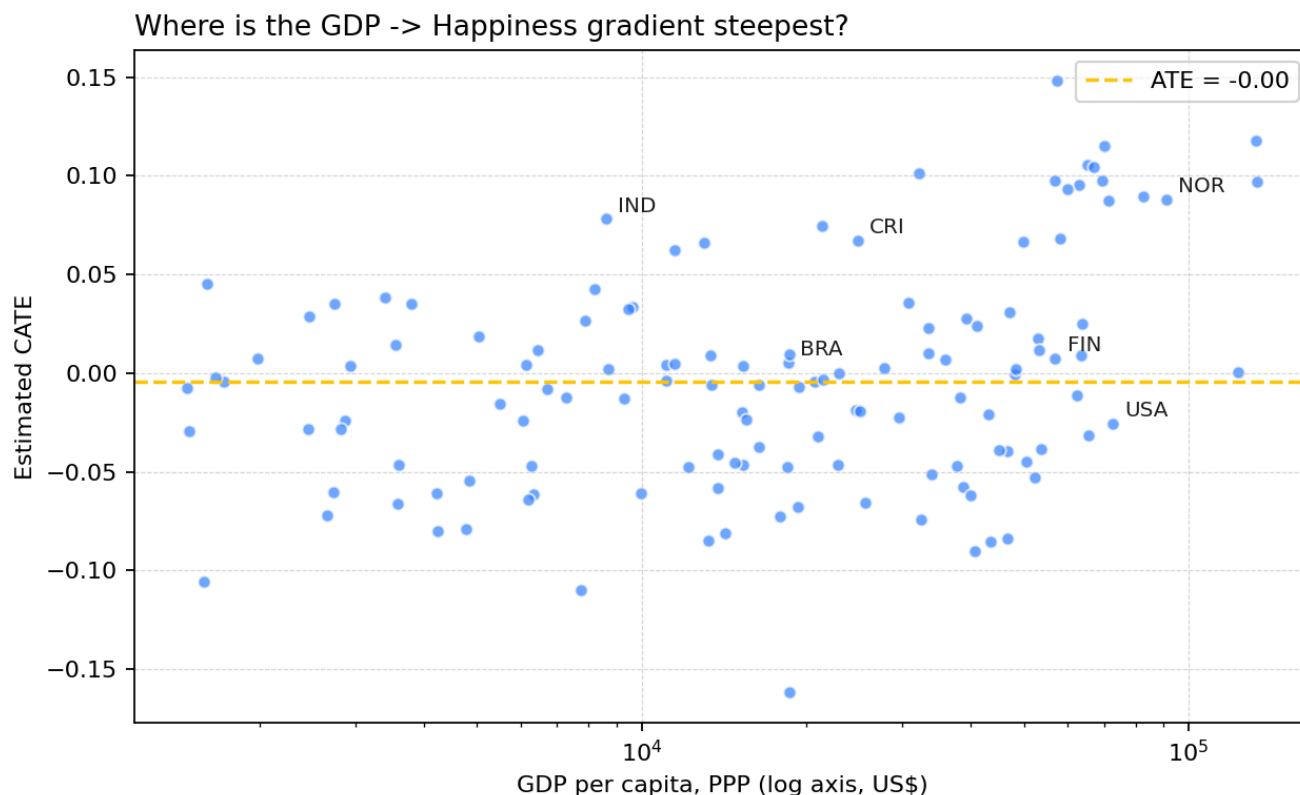
ATE = -0.004 (95% CI: [-0.365, +0.357])

CATEs: min = -0.16, median = -0.01, max = +0.15

The forest's average treatment effect lands very close to zero once we condition on the six moderators, far below the raw 0.80 OLS coefficient on log GDP. Almost all of the apparent income premium runs through those other covariates.

There *is* still genuine heterogeneity across countries, though. Which countries get the biggest happiness boost from being above the income median, and what predicts the size of that boost?

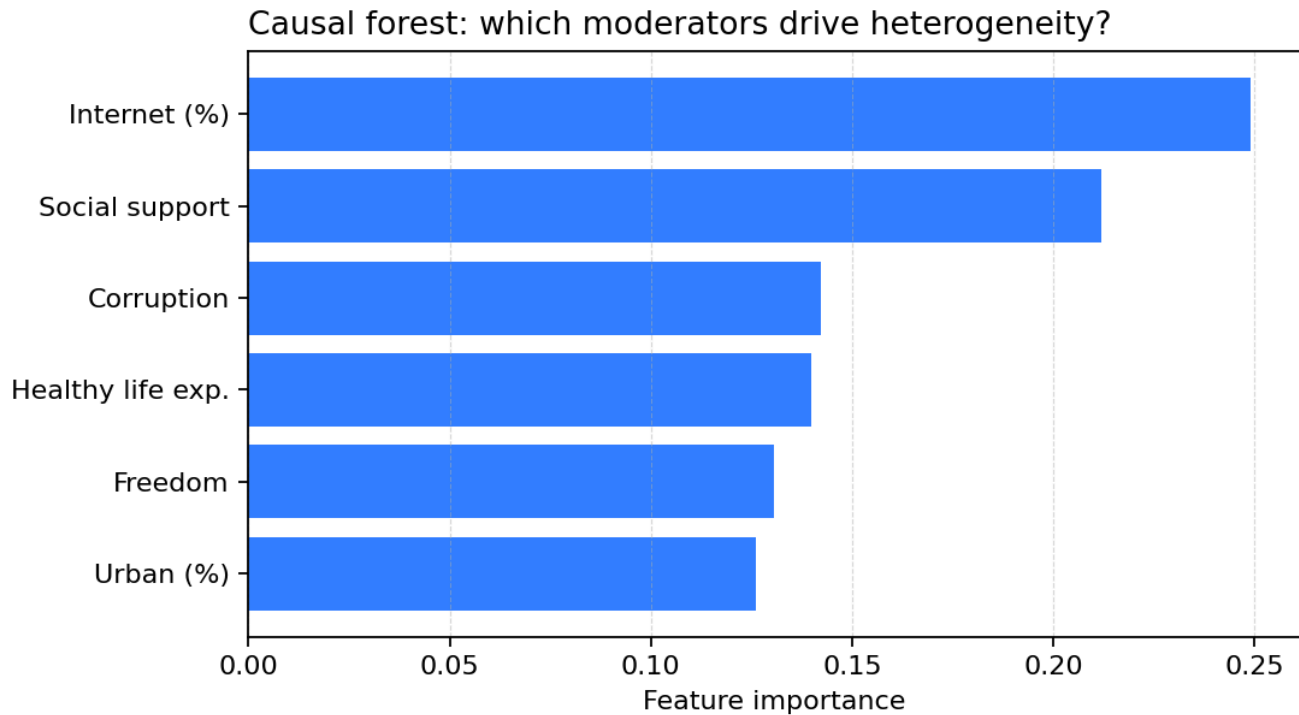
```
In [8]: display(Image(filename=FIG + "cate_by_gdp.png"))
```



Plotting each country's CATE against its income produces no clean monotone pattern. Some rich countries (Norway, Australia) come out modestly above the ATE line, others (the United States) sit close to or below zero. The poorest end is just as mixed: India is well *above* the average gradient while several sub-Saharan economies sit well below. Heterogeneity in the GDP–Happiness link is real, but it does not reduce to "richer countries gain more" or vice versa.

So what does explain the heterogeneity? The forest's split-importance gives us a direct answer.

```
In [9]: display(Image(filename=FIG + "cf_importance.png"))
```

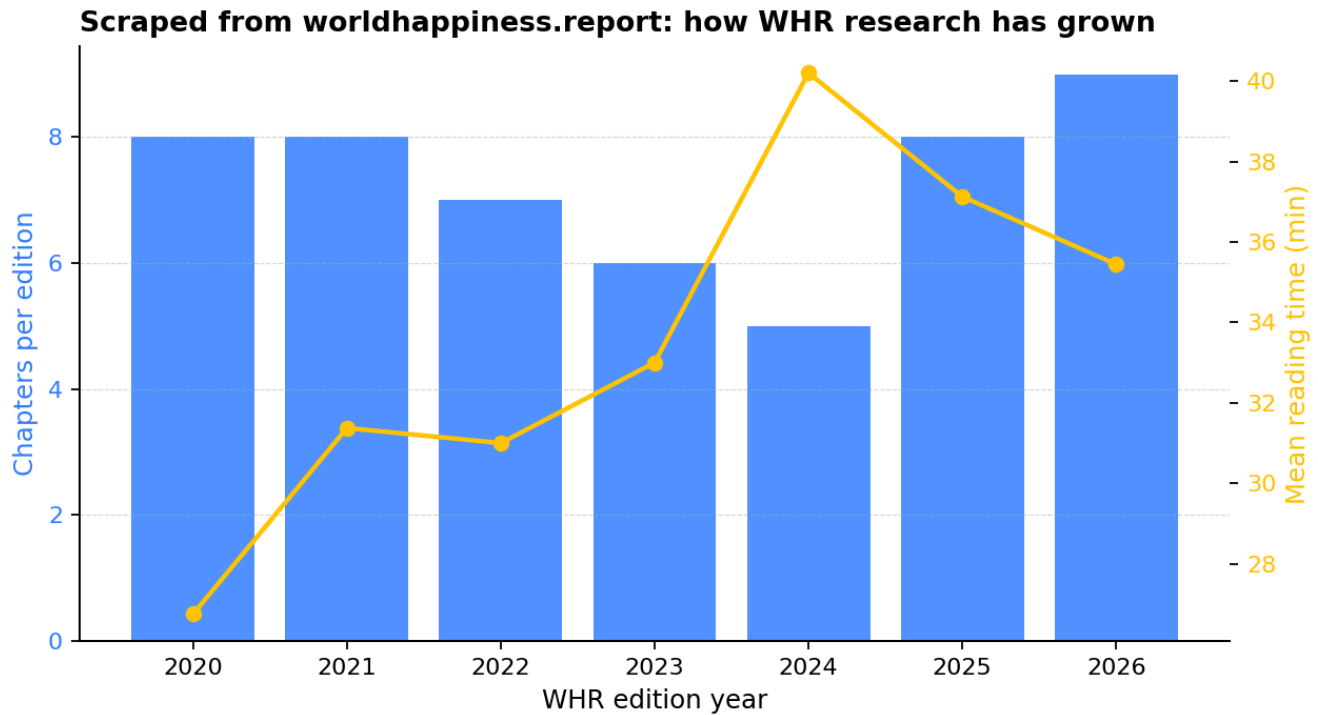



Internet penetration is the most important moderator, ahead of social support, corruption and the others. That fits a small but growing literature on digital connectivity and well-being: once we know whether a country's residents are *online*, the leftover role of GDP per capita shrinks a lot.

A bonus from the scrape: the Report itself

The rankings spreadsheet is the core of the analysis, but the chapter metadata I scraped from the seven editions has its own story. The crawler collected 51 chapters across 2020-2026 with author lists, affiliations and reading times. As far as I can tell that table does not exist anywhere else, and it lets us watch how a flagship interdisciplinary report has shifted over half a decade.

```
In [10]: display(Image(filename=FIG + "chapters_trend.png"))
```

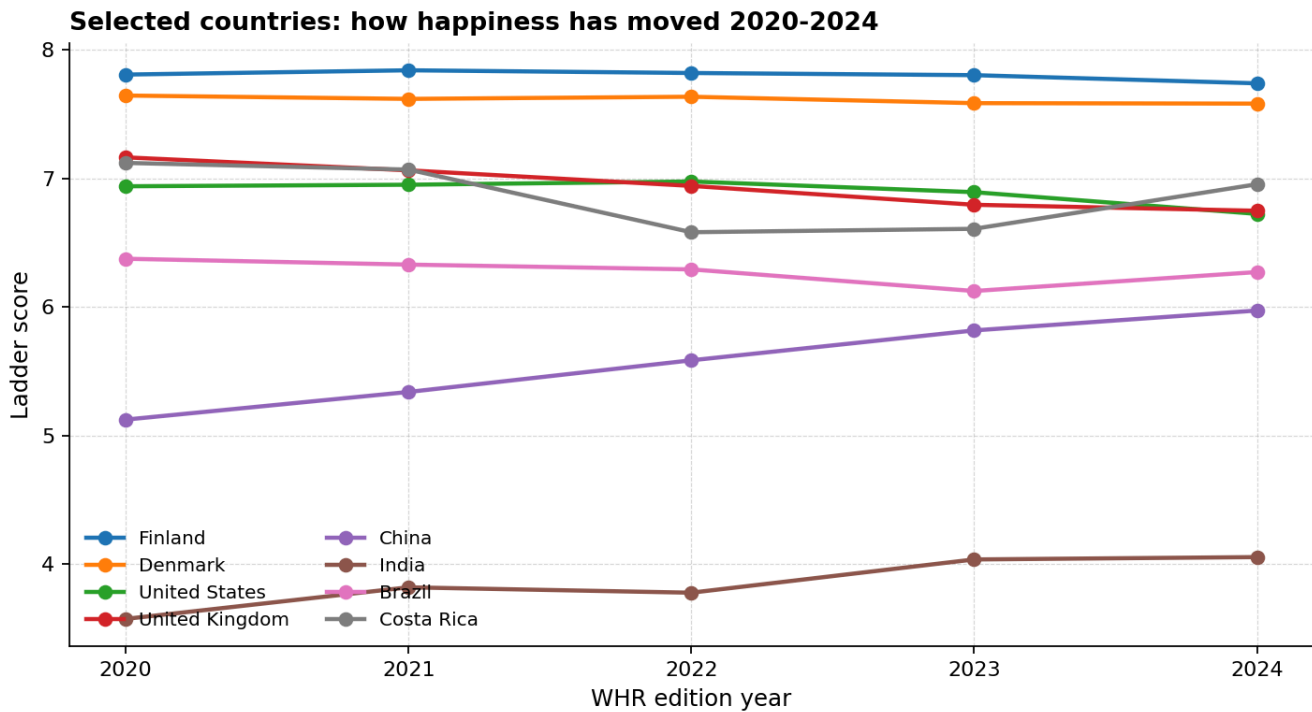


A couple of things to notice. The number of chapters per edition is volatile (2024 was unusually slim, with just five chapters), but average reading time has risen sharply, from about 27 minutes in 2020 to about 40 in 2024. The Report is, quite literally, a longer read. The affiliations field shows that John Helliwell, Jan-Emmanuel De Neve and Richard Layard have continued to anchor the editorial team, while recent editions have brought in a wider circle of specialists, especially social psychologists working on adolescents and digital well-being. The 2026 edition is more or less a thematic deep-dive on social media, and that pivot shows up clearly in the titles of the scraped chapter list.

Time series: where has happiness moved?

Because the scraper grabbed every annual `Figure_2.1.xls`, we can also compare country trajectories across editions. The plot is unremarkable at the very top (Finland is almost flat), but it does show the slow erosion of US scores relative to Northern Europe and a small post-pandemic recovery in parts of Latin America.

```
In [11]: display(Image(filename=FIG + "ladder_timeseries.png"))
```



Take-aways

A few things I take away from this exercise:

1. Money matters, but mostly *indirectly*. Once you control for social support, freedom, life expectancy and trust in institutions, the marginal contribution of income to subjective well-being is small.
2. The income-happiness link is genuinely heterogeneous, but not monotone. The causal forest produces CATEs roughly between -0.16 and $+0.15$, yet they do not line up cleanly with how rich a country is. Institutional quality and digital connectivity matter more than dollars.
3. Social capital is at least as potent as material wealth. In every Nordic country the biggest single contributor to the Ladder gap above the world mean is *social support*, not GDP.
4. Internet penetration is the dominant moderator. Split-importance ranks it ahead of social support, corruption, life expectancy, freedom and urbanisation, which lines up with the WHR 2026 edition's own pivot towards social-media research.
5. The Report itself is changing. Chapters have got longer and the 2026 edition pivots sharply onto social media; the scrape lets us watch that shift happen.

Replication

Everything is on GitHub. The repository contains a `Makefile` that rebuilds all eight figures and the regression table from scratch:

```
git clone https://github.com/maksimkitikov/Data-Science-in-Economics.git
cd Data-Science-in-Economics
pip install -r requirements.txt
make all
```

Sources & further reading

- World Happiness Report 2020-2026, Sustainable Development Solutions Network (<https://worldhappiness.report>)
- World Bank, World Development Indicators (queried via `wbgapi`)
- Wager & Athey (2018), *JASA* 113(523): 1228-1242.
- Athey, Tibshirani & Wager (2019), *Annals of Statistics* 47(2): 1148-1178.
- Davis & Heller (2017), *AEA P&P* — applied causal forests, recommended in the BEE2041 project brief.

Course materials referenced in the code

File	Where it shows up
Workflow, Modelling & Webscraping.pdf	scraper boilerplate, causal-forest recipe
Python for Data Management.pdf	merge with <code>validate=</code> and <code>indicator=</code>
Relational Database Management Systems.pdf	SQLite schema, JOIN query
Problem Set Solutions: Data Wrangling in Python.pdf	<code>wbgapi</code> usage
git & GitHub.pdf	branch / commit conventions
The Linux Command Line.pdf	<code>Makefile</code> build pipeline
Damian Clarke, BEE2041-2026 GitHub	code style, <code>immigrantEffects.py</code> , <code>scrape_xkcd_bs.py</code>